STRAIGHTENING LAWS IN ALGEBRAIC COMPLEXITY

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Abstract.

1 Introduction

⟨⟨ Note about how alot of material here is based on [DG25]. ⟩⟩

⟨⟨ I will flesh out the whole introduction more at a later date. ⟩⟩

Many problems in combinatorics, geometry, and computer science can be expressed in terms of the computation of some polynomials. For example, finding a perfect matching in a graph is equivalent to determining if it's *Tutte Matrix* has determinant identically zero [Tut47]. With this, we see that it is important to study the complexity of computing polynomials in general, and specifically computing special polynomials such as the determinant or permanent polynomials. Algebraic complexity theory is the study of the complexity of such computations.

⟨⟨ Talk a bit about the history of the field starting with Valient ⟩⟩

1.1 Lower Bounds and Polynomial Identity Testing **⟨⟨ Talk about Nisan-Wigderson and Kabanets-Impagliazzo ⟩⟩**

1.2 Geometric Complexity Theory **⟨⟨ Talk about Mulmuley-Sohoni ⟩⟩**

1.3 Straightening Laws **⟨⟨ Talk about Hodge, Young ⟩⟩**

2 Preliminaries on Algebraic Complexity

2.1 Computational Models

Definition 2.1 (Algebraic Circuits): An *algebraic circuit* over a field $\mathbb{F}$ is a directed acyclic graph whose *input nodes*, nodes of in-degree zero, are labeled either by variables $x_1, \dots, x_n$ or constants from $\mathbb{F}$. Nodes of in-degree greater than zero are labeled by Σ or Π which are *addition gates* or *multiplication gates* respectively. Each edge or *wire* is also labeled by a constant from $\mathbb{F}$.

Computation is defined inductively node by node. Input nodes compute their label. Every Σ gate computes the sum of their children multiplied by the constants along each wire, and computation at

Π gates is defined similarly. Nodes of out-degree zero are *outputs*. We say that an algebraic circuit *computes* $\{f_1(\overline{x}), \ldots, f_k(\overline{x})\}$ if these are the polynomials computed by all the outputs.

The *size* of a circuit is the total number of edges. Some sources also define the size to be the number of vertices. These are polynomially related as a directed simple graph with v vertices has $O(v^2)$ edges. It's *depth* is the length of the longest input-to-output path. The *fan-in* and *fan-out* of a node are in-degree and out-degree respectively. The *fan-in* and *fan-out* of the circuit is the maximum fan-in and fan-out respectively over all nodes in the circuit. The *degree* or *syntactic degree* of a circuit is the maximum degree of a polynomial computed by any of the nodes of the circuit.

Example 2.2: Let $f(x) = \sum_{i=0}^{d} a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

Then *Horner's Rule* gives a recurrence for efficiently computing $f(x)$:

$$p_0(x) = a_0, \qquad p_k(x) = a_{d-k} + x \cdot p_{k-1}(x).$$

This recurrence can be formulated as an algebraic circuit of size $O(d)$ and depth $O(d)$. In fact, this uses exactly n additions and n multiplications, which is optimal. The proof of optimality of the number of additions for computing a generic univariate polynomial $f(x)$ in this manner was given by [Ost54]. The corresponding proof of optimality for multiplications was given by [Pan66].

Definition 2.3 (Algebraic Formulas): An *algebraic formula* over a field $\mathbb{F}$ is an algebraic circuit whose underlying graph is a tree.

Definition 2.4 (Oracle Circuits and Formulas): For a fixed polynomial $f(\overline{x}) \in \mathbb{F}[\overline{x}]$, an *$f$-oracle circuit* is an algebraic circuit defined in the same manner as Definition 2.1, however instead of just Σ and Π gates we now additionally allow $f(\overline{x})$ gates. Note that such gates may have some inputs set as wires and some inputs set as fixed constants from $\mathbb{F}$. We also define an *f-oracle formula* as an f-oracle circuit whose underlying graph is a tree.

Definition 2.5 (Algebraic Branching Programs): An *algebraic branching program* (*ABP*) over a field $\mathbb{F}$ is a directed acyclic graph A such that:

- The nodes V of A are partitioned into non-empty subsets $V = V_0 \sqcup \cdots \sqcup V_d$ for some integer $d \geq 0$ and we call V_i the i-th layer. Layer V_0 has exactly one node called the *source* s and layer V_d has exactly one node called the *sink* t.
- Each edge in A goes from layer V_{i-1} to V_i for some $i \in [d]$.
- Each edge e in A is labeled with a polynomial $\gamma_e \in \mathbb{F}[\overline{x}]$ of degree at most one.

For a path $\overline{e} = (e_1, \ldots, e_d)$ from s to t, where each edge e_i goes from layer $i-1$ to layer i, define the polynomial $\gamma_{\overline{e}} := \prod_{i=1}^{d} \gamma_{e_i}$. Then A computes the polynomial $\sum_{\substack{\text{path } \overline{e} \\ \text{from } s \text{ to } t}} \gamma_{\overline{e}}$.

The *size* of an ABP A is the number of edges in A. Some sources also define the size to be the number of vertices. These are polynomially related as a directed simple graph with v vertices has has $O(v^2)$ edges. The *width* of A is the maximum number of vertices in any layer of A. The *depth* of A is the number of layers.

Remark 2.6: Definition 2.5 is really the definition of a *layered* ABP, where each edge only goes from one layer to the next without skipping any layers. However, one can convert a non-layered ABP to a layered ABP with only a quadratic blowup in size.

Definition 2.7 (Determinantal Complexity): For a polynomial $f(\overline{x})$ over a field $\mathbb{F}$, the *determinantal complexity* of $f(\overline{x})$, denoted $\mathrm{dc}(f)$, is the smallest m such that there exists an $m \times m$ matrix M with entries in $\mathbb{F}[\overline{x}]_{\leq 1}$ such that $\det_m(M) = f(\overline{x})$.

Lemma 2.8 ([Val79; Bür24]): Suppose that $f(\overline{x})$ can be computed by an algebraic branching program of size m. Then $\mathrm{dc}(f) \leq m$.

2.2 Basic Structural Results

Lemma 2.9: **⟨⟨ Depth reduction, conversions from circuits to formulas / ABPs and vice versa. ⟩⟩**

Lemma 2.10: **⟨⟨ Elimination of Division ⟩⟩**

Definition 2.11: For $f \in \mathbb{F}[\overline{x}, t]$ and an integer $i \in \mathbb{Z}_{\geq 0}$, denote by $\mathrm{coeff}_{t^i}(f)$ the coefficient of t^i in f when f is viewed as a univariate polynomial in t over the ring $\mathbb{F}[\overline{x}]$.

Lemma 2.12 ([DG25, Lemma 2.29]): Let $g \in \mathbb{F}[\overline{x}, t]$, and let $f \in \mathbb{F}[\overline{x}, t]$ be a degree-d polynomial such that for each $\alpha \in \mathbb{F}$, the polynomial $f(\overline{x}, \alpha)$ is computed by a g-oracle circuit C_α of size at most s and depth at most Δ. Assume that $|\mathbb{F}| \geq d+1$. Then, for every $0 \leq i \leq d$, there exists a g-oracle circuit D_i of size $O(ds)$ and depth $\Delta + 1$ that computes $\mathrm{coeff}_{t^i}(f)$, with its top gate being an addition gate and bottom Δ layers consisting of $d+1$ copies of g-oracle circuits of the form C_α. Moreover, if the top gate of each C_α is an addition gate, then the depth bound can be improved to Δ.

Proof: By definition,

$$f = \sum_{i=0}^{d} \mathrm{coeff}_{t^i}(f) t^i.$$

Let $\alpha_1, \ldots, \alpha_{d+1}$ be $d+1$ distinct elements in $\mathbb{F}$. Then we have that

$$\begin{pmatrix} \alpha_1^d & \alpha_1^{d-1} & \cdots & \alpha_1 & 1 \\ \alpha_2^d & \alpha_2^{d-1} & \cdots & \alpha_2 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_d^d & \alpha_d^{d-1} & \cdots & \alpha_d & 1 \\ \alpha_{d+1}^d & \alpha_{d+1}^{d-1} & \cdots & \alpha_{d+1} & 1 \end{pmatrix} \begin{pmatrix} \mathrm{coeff}_{t^d}(f) \\ \mathrm{coeff}_{t^{d-1}}(f) \\ \vdots \\ \mathrm{coeff}_{t}(f) \\ \mathrm{coeff}_{1}(f) \end{pmatrix} = \begin{pmatrix} f(\overline{x}, \alpha_1) \\ f(\overline{x}, \alpha_2) \\ \vdots \\ f(\overline{x}, \alpha_d) \\ f(\overline{x}, \alpha_{d+1}) \end{pmatrix}$$

The $(d+1) \times (d+1)$ matrix $(\alpha_i^{d+1-j}) \in \mathbb{F}^{(d+1)\times(d+1)}$ is a Vandermonde matrix. It's determinant is $\prod_{i<j}(\alpha_j - \alpha_j)$. As all the α_i are distinct, this determinant is nonzero so that the matrix is invertible. Thus, there exists a $(d+1) \times (d+1)$ matrix $(z_{i,j}) \in \mathbb{F}^{(d+1)\times(d+1)}$ such that

$$\begin{pmatrix} \mathrm{coeff}_{t^d}(f) \\ \mathrm{coeff}_{t^{d-1}}(f) \\ \vdots \\ \mathrm{coeff}_{t}(f) \\ \mathrm{coeff}_{1}(f) \end{pmatrix} = \begin{pmatrix} z_{1,1} & z_{1,2} & \cdots & z_{1,d} & z_{1,d+1} \\ z_{2,1} & z_{2,2} & \cdots & z_{2,d} & z_{2,d+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{d,1} & z_{d,2} & \cdots & z_{d,d} & z_{d,d+1} \\ z_{d+1,1} & z_{d+1,2} & \cdots & z_{d+1,d} & z_{d+1,d+1} \end{pmatrix} \begin{pmatrix} f(\overline{x}, \alpha_1) \\ f(\overline{x}, \alpha_2) \\ \vdots \\ f(\overline{x}, \alpha_d) \\ f(\overline{x}, \alpha_{d+1}) \end{pmatrix}$$

Fix $0 \le i \le d$. We now have an explicit expression for $\mathrm{coeff}_{t^i}(f)$:

$$\mathrm{coeff}_{t^i}(f) = \sum_{j=1}^{d+1} z_{d+1-i,j} f(\overline{x}, \alpha_j) \tag{1}$$

With Equation (1), we can now describe the circuit for $\mathrm{coeff}_{t^i}(f)$. The scalars $\alpha_1, \ldots, \alpha_{d+1}$ are fixed a priori, and hence so are the $z_{i,j}$ for $i, j \in [d+1]$. Recall that scalar multiplications along wires are free. For each $j \in [d+1]$, computing $f(\overline{x}, \alpha_j)$ requires a single g-oracle circuit C_{α_j} of size at most s and depth Δ. Hence, evaluating all $d+1$ such terms requires total size $O(ds)$. The top-level summation uses one addition gate and $d+1$ scalar multiplications by the $z_{d+1-i,j}$ along $d+1$ wires. Therefore, Equation (1) yields a g-oracle circuit D_i of size $O(ds+d) = O(ds)$ and depth $\Delta + 1$ computing $\mathrm{coeff}_{t^i}(f)$. Moreover, if the top gate of each C_{α_j} is an addition gate, then the top two layers of D_i consist entirely of additions and can be merged to reduce the depth bound to Δ. □

Remark 2.13: The condition $|\mathbb{F}| \ge d+1$ in Lemma 2.12 is required for interpolation. We note that if g is given not as a black box but as a white-box circuit, then its homogeneous components can be extracted in a gate-by-gate fashion [Str73], which requires no lower bound on $|\mathbb{F}|$.

Lemma 2.14 ([KS01, Lemma 4]): Let C be any collection of distinct linear forms in variables $z_1, \ldots, z_\ell$ with coefficients in the range $\{0, \ldots, K\}$ for some integer $K \in \mathbb{Z}_{\ge 0}$. Let $\varepsilon > 0$. Let $z_1, \ldots, z_\ell$ be independently and uniformly chosen from $\{0, \ldots, M\}$ at random, where $M \ge K\ell/\varepsilon$. Then, with probability at least $1 - \varepsilon$, there is a unique form in C of minimum value at $z_1, \ldots, z_\ell$.

A simple rephrasing of Lemma 2.14 allows us to isolate a term of a multivariate polynomial $f(y_1,\ldots,y_\ell)$ under a random monomial substitution $y_i \mapsto w^{z_i}$:

Corollary 2.15 ([DG25, Corollary 2.27]): Let $K \in \mathbb{Z}_{\geq 0}$. Let A be any ring, and let $f(y_1,\ldots,y_\ell) \in A[\overline{y}]$ be a polynomial whose individual degree in each variable is at most K, i.e., $\deg_{y_i}(f) \leq K$ for all $i \in [\ell]$. Fix $\varepsilon > 0$. Choose $z_1,\ldots,z_\ell$ independently and uniformly at random from $\{0,\ldots,M\}$, where $M \geq K\ell/\varepsilon$. Let w be a new indeterminate, and define $\phi : A[y_1,\ldots,y_\ell] \to A[w]$ to be the ring homomorphism sending $y_i \mapsto w^{z_i}$ for each $i \in [\ell]$. Then, with probability at least $1-\varepsilon$, there exists a unique monomial $\mathfrak{m} = y_1^{e_1}\cdots y_\ell^{e_\ell}$ among all monomials of f such that $\phi(\mathfrak{m}) = w^{e_1 z_1 + \cdots + e_\ell z_\ell}$ attains the minimum degree in w.

2.3 Border Complexity

Definition 2.16: We say that $g \in \mathbb{F}[x_1,\ldots,x_n]$ is *border computed* by a circuit C over $\mathbb{F}(\varepsilon)$ if C computes some polynomial $h \in \mathbb{F}(\varepsilon)[x_1,\ldots,x_n]$, and $h - g \in \varepsilon \cdot \mathbb{F}[[\varepsilon]][x_1,\ldots,x_n]$, where $\mathbb{F}[[\varepsilon]]$ denotes the ring of formal power series in ε.

In the settings where $\mathbb{F}[\varepsilon][x_1,\ldots,x_n]$ can be equipped with the Euclidean topology in a natural way, such as $\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$, this would imply that $h \to g$ as $\varepsilon \mapsto 0$. However, this does not mean that we could just substitute ε by zero in C and obtain a circuit exactly computing g since C may use scalars in $\mathbb{F}(\varepsilon)$, such as $1/\varepsilon$, that are not well-defined at $\varepsilon = 0$.

Example 2.17 ([DL25]): In order to interpolate the coefficients of $f(\overline{x}, t) = \sum_{i=0}^{d} \mathrm{coeff}_{t^i}(f) \cdot t^i$ over $\mathbb{F}$ as in Lemma 2.12, one needs that $|\mathbb{F}| \geq d+1$. However, we can border compute $\mathrm{coeff}_{t^r}(f)$ if $|\mathbb{F}| \geq r+1$.

Suppose $\mathbb{F}$ has distinct elements $\{\alpha_0,\ldots,\alpha_r\}$. Then, there exists elements $\beta_0,\ldots,\beta_r \in \mathbb{F}$ such that

$$\mathrm{coeff}_{t^r}(f) = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r} \beta_i \cdot f(\overline{x}, \varepsilon\alpha_i)\right).$$

Indeed let $g(\overline{x}, t) = \sum_{i=0}^{r} \mathrm{coeff}_{t^i}(f) \cdot t^i$ and $h(\overline{x}, t) = f(\overline{c}, t) - g(\overline{x}, t)$. Then standard interpolation on $\varepsilon\alpha_i$ gives us that there exists field constants $\beta_0,\ldots,\beta_r$ such that $\varepsilon^r \mathrm{coeff}_{t^r}(f) = \sum_{i=0}^{r} \beta_i \cdot g(\varepsilon\alpha_i)$. Thus, we have that

$$\begin{aligned}
&\lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r} \beta_i \cdot f(\overline{x}, \varepsilon\alpha_i)\right) \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r} \beta_i \cdot g(\overline{x}, \varepsilon\alpha_i)\right) + \lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r} \beta_i \cdot h(\overline{x}, \varepsilon\alpha_i)\right) \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r} \beta_i \cdot g(\overline{x}, \varepsilon\alpha_i)\right) + 0 = \mathrm{coeff}_{t^r}(f).
\end{aligned}$$

3 Preliminaries on Straightening Laws

3.1 Determinantal Ideals

3.2 Pfaffian Ideals

3.3 Hankel Ideals

3.4 Symmetric Determinantal Ideals ⟨⟨ Only if time ⟩⟩

3.5 Permanental Ideals ⟨⟨ Only if time ⟩⟩

4 Prior Work

4.1 Generalities on Complexity in Ideals Roughly speaking, the complexity of an ideal I under a given algebraic model (such as algebraic circuits or algebraic branching programs) is defined as the minimum complexity, in that model, of any nonzero polynomial $f \in I$ [Gro20]. Proving lower bounds for ideals is harder than for individual polynomials, since one must establish a bound for every nonzero $f \in I$ rather than just a single polynomial. This motivates the search for a property of the following form: if an arbitrary nonzero $f \in I$ is efficiently computable in a given model, then some polynomial from a distinguished set $S = \{g_1, \ldots, g_k\}$ (often a natural generating set of I or some related set with desirable properties) is also efficiently computable in the given model. In this way, proving lower bounds for arbitrary nonzero $f \in I$ reduces to proving them for $g_1, \ldots, g_k$. We call such a property a *closure result*, by analogy with closure under factorization in the principal ideal case. In short, closure results reduce the task of showing hardness for all polynomials in I to proving hardness for a small set of distinguished polynomials.

Definition 4.1 (Hitting Set Generator (HSG)):

Definition 4.2 (Polynomial Identity Testing (PIT)):

Closure results for ideals are useful for constructing hitting set generators, which are the algebraic-complexity analogs of pseudorandom generators and can be used to derandomize polynomial identity testing (PIT). The idea is as follows: for $s < n$, we seek a hitting set generator $G \colon \mathbb{F}^s \to \mathbb{F}^n$ against a family $\mathcal{C}$ of n-variate polynomials. This means that G has the property that for any nonzero polynomial $f \in \mathcal{C}$, we have $f \circ G \neq 0$. Here, one should think of $\mathcal{C}$ as a family of "low-complexity" polynomials. Let I be the ideal of polynomials f satisfying $f \circ G = 0$, also known as the *vanishing ideal* of G [HMM24]. To show that G has the desired hitting set property, it suffices to show that every nonzero $f \in I$ has sufficiently high complexity so that no such f can belong to $\mathcal{C}$. With a closure result for I, this reduces to proving lower bounds for a distinguished set of polynomials $\{g_1, \ldots, g_k\}$ for I. This can be viewed as a particular way of implementing the "hardness versus randomness" paradigm of [NW94; KI04] in the setting of algebraic complexity.

The general question we want to study is of the following form:

Question 4.3: Say that $I = \langle g_1, \ldots, g_k \rangle$. Let $0 \neq f \in I$. How does the complexity of f compare to the complexity of the generators $g_1, \ldots, g_k$.

Note that this question as stated is not well formed. **⟨⟨ Elaborate: ⟩⟩** $\langle \det, \det + \mathrm{perm} \rangle$

4.1.1 Principal Ideals

Let us consider the special case of Question 4.3 where $I = \langle g \rangle$ is a *principal ideal*. Then our question is now the following:

Question 4.4: Let $0 \neq f \in \langle g \rangle$. Does g have a small f-oracle circuit?

The affirmative was actually conjectured by Bürgisser:

Conjecture 4.5 ([Bür00, Conjecture 8.3]): If g is a factor of $f \neq 0$, then $\mathrm{size}(g) \leq \mathrm{poly}(\mathrm{size}(f), \deg(g))$.

There has been much progress on Conjecture 4.5. Factoring in algebraic complexity is well studied, see the survey of Bhargav, Dwivedi, and Saxena for an overview of recent progress [BDS26]. For example, it is known that for fields of characteristic zero that $\mathrm{size}(g) \leq \mathrm{poly}(\mathrm{size}(f), \deg(f))$ [Kal87; Bür00].

4.1.2 More Than One Generator

5 Current Work on Ideals with Straightening Laws

5.1 Approximative Results

5.2 Debordering

6 Future Work and Open Problems

6.1 Symmetric Determinantal Ideals

6.2 Permanental Ideals

6.3 Hankel and Toeplitz Ideals

6.4 Complexity of Straightening

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